

Prarthana Krishnamurthy

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Summary

Machine Learning Engineer with 4+ years of experience building predictive and generative models alongside the production data pipelines that feed them. Delivers models end to end: PyTorch and scikit-learn training, FastAPI and MCP serving, PySpark ETL, and Docker with CI/CD deployment.

Experience

Machine Learning Intern

May 2026 – Jul 2026

Alkermes Inc.

Waltham, MA

- Built supervised models (PyTorch, scikit-learn) predicting Chemical properties at R^2 0.7 and Spearman 0.73
- Deployed a FastAPI serving layer for the trained models, cutting inference latency 2.5x
- Exposed the model endpoints through an MCP server, letting AI agents invoke them as callable tools
- Established the team's ML deployment infrastructure (Docker, GitLab CI/CD), enabling reliable model releases
- Applied SHAP to translate black-box model outputs into explanations for non-technical stakeholders

Machine Learning Research Assistant

Jan 2025 – Apr 2026

Northeastern University (Advisor: Prof. Dino Konstantopoulos)

Boston, MA

- Architected transformer and diffusion models using flow matching for non-autoregressive text generation
- Processed 30GB+ raw text corpora with spaCy into training-ready NLP datasets for model development
- Optimized DataLoader memory footprints by dynamically extracting frozen BERT embeddings under contexts, preventing autograd accumulation and OOM errors during large batch generation
- Scaled training across 8x NVIDIA H100 GPUs via PyTorch FSDP, sustaining 95%+ GPU utilization
- Resolved sampling error accumulation by implementing DPM-Solver++, reducing discretization error to 0.65
- Delivered SOTA metrics including BLEU-4: 0.267 and ROUGE-L: 0.742, demonstrating end-to-end ownership

Software Engineer

Sep 2021 – Dec 2023

Societe Generale Global Solution Centre

Bangalore, KA

- Architected an AWS data lake supporting 1M+ daily records, designing ingestion and partitioning strategies
- Productionized 3+ Dockerized ML applications on Azure Kubernetes Service, processing 1M+ daily transactions
- Improved fraud detection precision 18% at fixed recall with transaction-velocity and NetworkX graph features
- Built data drift detection and automated retraining triggers on transaction feature distributions, reducing model degradation incidents by 30% across quarterly retraining cycles
- Engineered PySpark pipelines for raw-to-silver processing, enforcing schemas and deduplicating records

Software Engineering Intern

Mar 2021 – May 2021

Capgemini

Bangalore, KA

- Built 15+ RESTful microservices (Java 17, Spring Boot), cutting API latency 35% with JPA caching
- Implemented multi-tenant microservices, ensuring high availability and data consistency for enterprise clients
- Engineered unit and integration test suites (JUnit, Mockito) to catch regressions and ensure API reliability
- Developed a client-facing React UI over those microservices, exposing services and data directly to clients

Publications

- [Non-Autoregressive Text Generation Using Diffusion Models](#), Master's Thesis, Northeastern University

Technical Skills

Programming Languages: Python (NumPy, SciPy, Pandas, Matplotlib), SQL, Java, React, Bash, Linux

ML/DL Frameworks: PyTorch (FSDP, DDP), TensorFlow, Keras, scikit-learn, XGBoost, LightGBM

Deep Learning & Generative AI: Transformers, Diffusion Models, Flow Matching, Optuna, LLMs, RAG

Cloud, MLOps & Deployment: AWS (S3, ECS, Lambda, Kinesis, DynamoDB, CloudWatch), Azure (Data Lake, AKS), Docker, Kubernetes, Terraform, FastAPI, MCP, REST APIs, Microservices, Git, CI/CD

Data Engineering: Airflow, PySpark, Elasticsearch, ETL/ELT Pipelines, Parquet, Kafka, Apache Flink, Hadoop

Mathematical Foundations: Stochastic Calculus, Linear Algebra, or Optimization.

Education

Northeastern University, MS in Computer Software Engineering (GPA: 3.84/4.0)

Jan 2024 – Dec 2025

Visvesvaraya Technological University, BE in Electronics and Communication

Aug 2017 – Aug 2021